DMET1001 – Image Processing

Assignment 2 – Semantic Segmentation

---

1. Introduction

The objective of this assignment is to investigate the effect of different loss functions on semantic segmentation performance using a lightweight U-Net architecture. Four different models were trained using different loss functions and evaluated using Dice Score, Intersection over Union (IoU), and HD95 metrics.

The implemented models are:

- Model A using Dice Loss
- Model B using Hausdorff-inspired Loss
- Model C using Dice + BCE Loss
- Model D using a Combined Loss

The goal of this comparison is to analyze the strengths and weaknesses of overlap-based losses and boundary-aware losses in terms of segmentation accuracy, boundary quality, and training stability.

---

2. Dataset Description & Preprocessing

The Oxford-IIIT Pet Dataset was used for this assignment. The dataset contains approximately 7,400 pet images with trimap segmentation masks for 37 pet breeds. A subset of 1000 image-mask pairs was sampled using a fixed random seed (`seed = 42`) to ensure reproducibility.

<u>The same split was used across all models:</u>

- Training set: 70% (700 images)
- Validation set: 15% (150 images)
- Test set: 15% (150 images)

<u>The trimap masks contain three labels:</u>

- 1 → foreground (pet)
- 2 → background
- 3 → uncertain border region

The masks were converted into binary masks by treating only label `1` as foreground and all other labels as background.

Preprocessing Steps:

- Resize images and masks to `256 × 256`
- Convert images to grayscale
- Normalize image pixel values to `[0,1]`
- Apply binary segmentation masks
- Data augmentation on training set only:
  - Random horizontal flip
  - Random rotation within ±15°

The same augmentation was applied to both image and mask to preserve alignment.

---

3. Segmentation Model Architecture

A small U-Net architecture was used for all experiments.

Architecture Details

The model consists of:

- Encoder feature depths: `[32, 64, 128, 256]`
- Double convolution blocks with:
  - Convolution
  - Batch Normalization
  - ReLU activation
- Max-pooling in encoder
- Transposed convolution upsampling in decoder
- Skip connections between encoder and decoder
- Final `1×1` convolution followed by sigmoid activation

The model outputs a binary segmentation mask.

Total trainable parameters:

- `7,762,465`

The same architecture was used for all models to ensure fair comparison between loss functions.

---

4. Loss Functions

4.1 Dice Loss (Model A)

Dice Loss measures overlap between predicted masks and ground truth masks.

The Dice Score is defined as:

Dice Score = (2 × Intersection) / (Prediction + Ground Truth)

Dice Loss is calculated as:

Dice Loss = 1 − Dice Score

Dice Loss is commonly used in segmentation because it focuses on overlap quality and handles class imbalance better than standard pixel-wise losses.

Observed Behavior:

Model A achieved:

- Best Dice Score
- Best IoU
- Strong overall segmentation quality

This happened because Dice Loss directly optimizes overlap between predicted and target regions.

4.2 Hausdorff-inspired Loss (Model B)

The Hausdorff-inspired loss is a differentiable approximation of the HD95 metric.

The classical Hausdorff Distance measures the maximum boundary deviation between prediction and ground truth masks. However, it is not differentiable and therefore unsuitable for gradient-based optimization.

Instead, this implementation uses precomputed distance transform maps for foreground and background regions. Prediction errors are weighted according to their distance from object boundaries.

The loss uses:

- Foreground distance transform
- Background distance transform
- Softplus function for smooth gradients

This loss focuses on boundary accuracy rather than only region overlap.

Advantages:

- Improves boundary accuracy
- Penalizes spatial boundary errors
- Encourages sharper segmentation edges

Observed Behavior:

Model B achieved:

- Best HD95 score
- Better boundary alignment than other models

Test HD95:

- `38.94` pixels (lowest among all models)

This happened because the distance-map weighting penalizes spatial boundary errors more strongly, forcing the model to improve contour accuracy.

However, Dice and IoU scores were slightly lower than Model A because the optimization focused more on boundaries than complete region overlap.

4.3 Tversky Loss (Model C)

Tversky Loss is a generalized form of Dice Loss that separately weights false positives and false negatives.

The model used:

- $\alpha = 0.3$
- $\beta = 0.7$

This configuration penalizes false negatives more heavily.

Reason for Choosing Tversky Loss

The loss was selected because segmentation masks may contain small or thin foreground regions, and false negatives are important to reduce in segmentation tasks.

Increasing β encourages the model to preserve foreground regions more aggressively.

Comparison with Dice and Hausdorff Loss

Compared to Dice Loss:

- Tversky provides more control over FP/FN penalties
- Better sensitivity toward missing foreground pixels

Compared to Hausdorff-inspired Loss:

- Tversky still focuses mainly on overlap
- Does not explicitly optimize boundaries

Observed Behavior:

Model C achieved acceptable overlap performance but weaker boundary accuracy.

Results:

- Dice: `0.8327`
- IoU: `0.7155`
- HD95: `48.37`

The higher HD95 indicates weaker boundary localization compared to Model B.

4.4 Combined Loss (Model D)

The assignment required combining the two best-performing losses from Models A–C.

Combined Loss Formula:

Lcombined = αL1 + βL2

Selected losses:

- A_Dice (Dice Score: `0.8466`)
- C_Tversky (Dice Score: `0.8397`)

These were selected because they achieved the highest validation Dice scores.

Selected coefficients:

- α = 0.5
- β = 0.5

The coefficients were selected based on validation Dice ranking. Since both losses achieved similar Dice performance, equal weights were used.

Observed Behavior:

The combined model produced balanced performance:

- Dice: `0.8452`
- IoU: `0.7340`
- HD95: `44.31`

The combined loss improved stability and maintained strong overlap performance, but it did not outperform Model A in Dice/IoU or Model B in HD95.

This happened because both selected losses are overlap-focused losses and do not explicitly optimize boundary precision.

---

5. Training Configuration

The models were trained using the following configuration:

Parameter	Value
Framework	PyTorch
Optimizer	Adam
Learning Rate	0.001
Batch Size	8
Epochs	60
Early Stopping	True
Random Seed	42

All models were trained until performance saturated (converged)

Training was performed using: GPU T4 x2

---

6. Evaluation Metrics

The models were evaluated using:

- Dice Score
- Intersection over Union (IoU)

- HD95

<u>Test Set Results:</u>

```
============================================================
Model                        Dice      IoU       HD95
============================================================
Model A (Dice)             0.8531   0.7455      40.22
Model B (Hausdorff)        0.8484   0.7388      38.94
Model C (Tversky)          0.8327   0.7155      48.37
Model D (Combined)         0.8452   0.7340      44.31
============================================================
Best Dice:  Model A (Dice)
Best IoU:   Model A (Dice)
Best HD95:  Model B (Hausdorff)
```

---

7. Training Behavior and Convergence Analysis

<u>Loss Curves Analysis:</u>

The training and validation loss curves show that all models converged successfully.

<u>Model A — Dice</u>

- Smooth convergence after early fluctuations
- Validation spikes around epoch 15
- Final training and validation losses remained close

The temporary instability likely happened because Dice Loss is highly sensitive to overlap mismatch during early segmentation learning.

<u>Model B — Hausdorff-inspired</u>

- Stable decreasing trend
- Lower learning rate (`5e-4`) improved optimization stability
- Small validation oscillations throughout training

The oscillations occurred because distance-transform weighting creates stronger gradients near boundaries.

Model C — Tversky

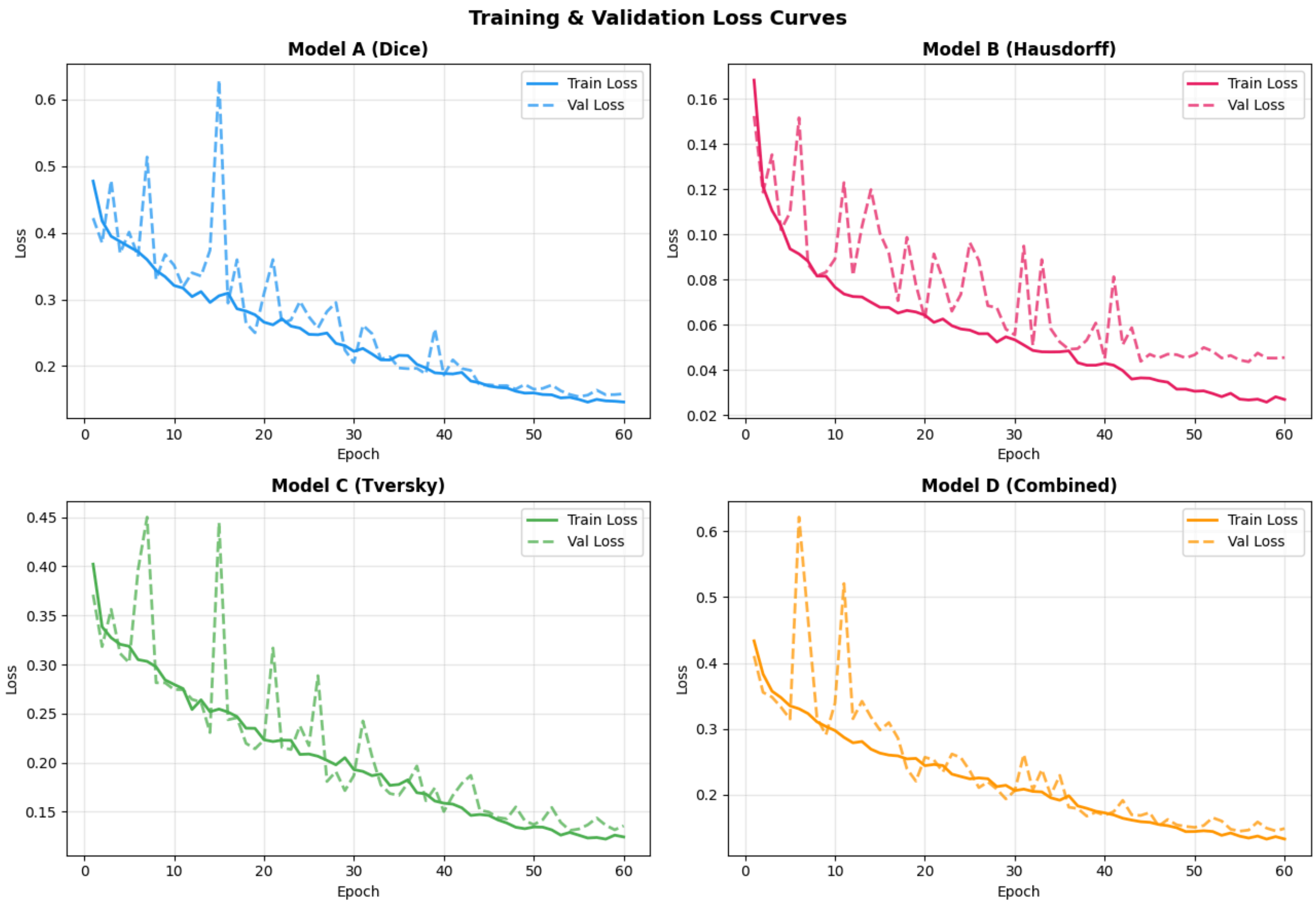
- Converged steadily
- Moderate instability in validation loss
- More fluctuations than Dice Loss

This occurred because stronger false-negative weighting changes optimization balance during training.

Model D — Combined

- Stable convergence overall
- Slower improvement during early epochs
- Final validation performance close to Model A

The slower convergence occurred because the optimizer had to balance two overlap-based objectives simultaneously.



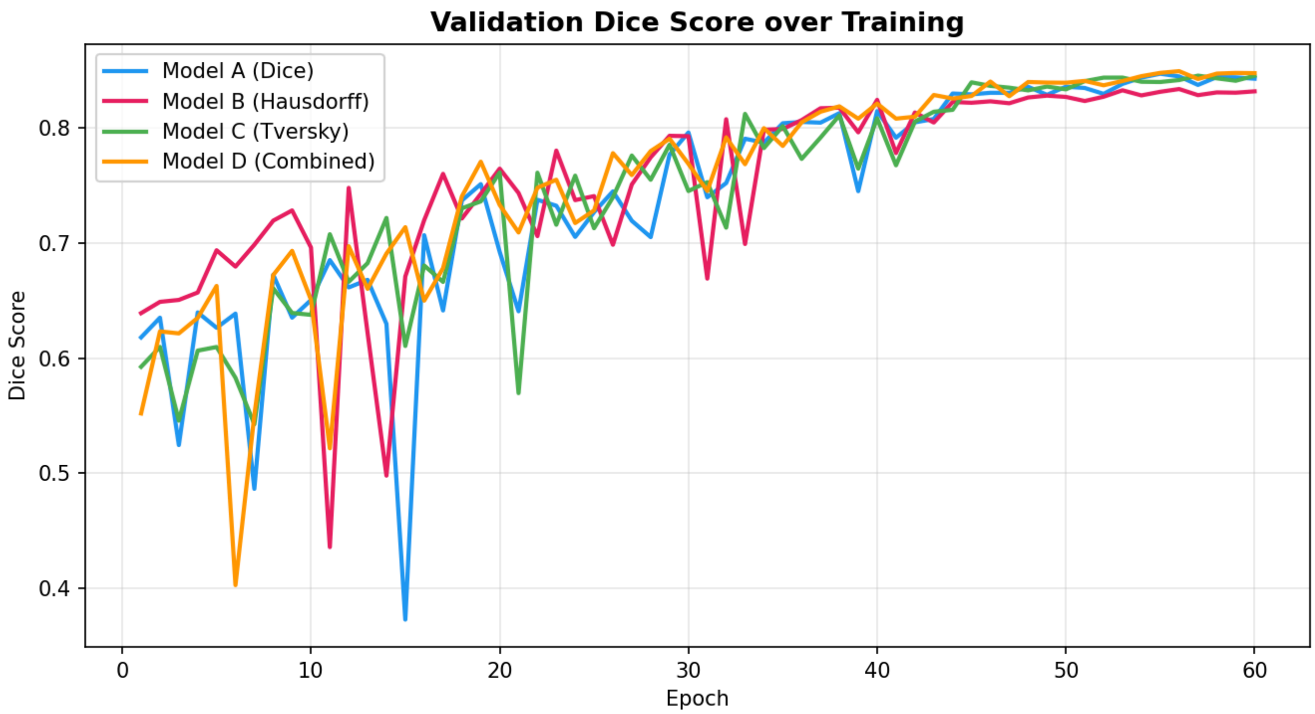
Validation Dice Curves:

The validation Dice curves show gradual improvement for all models.

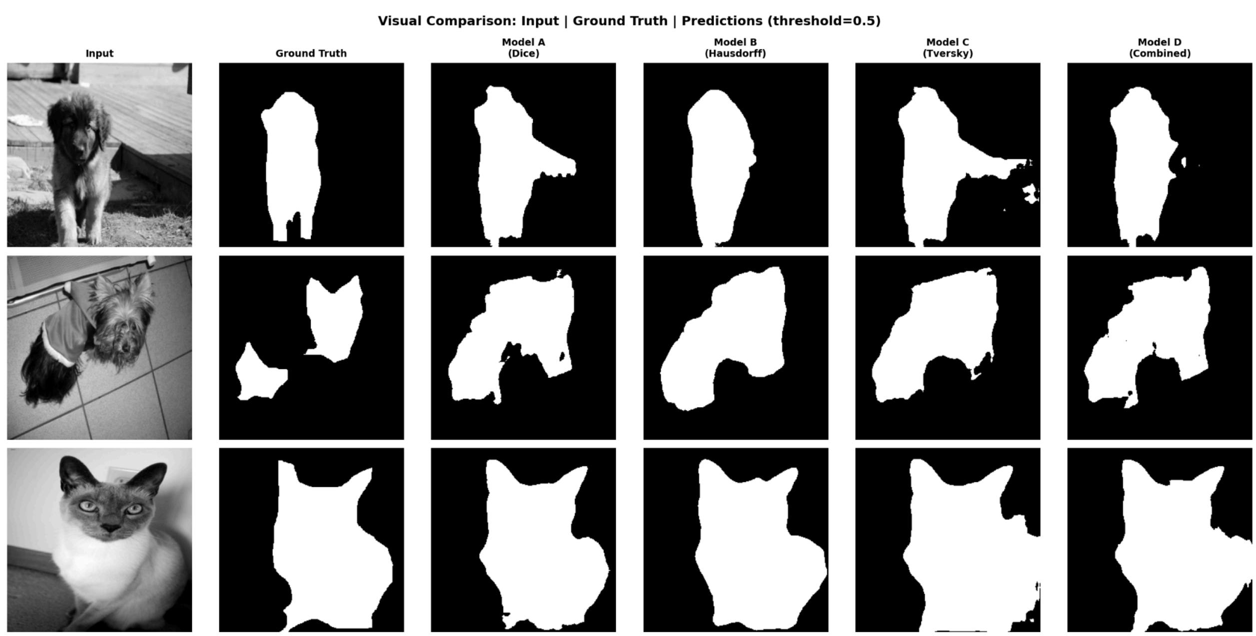
- Model A achieved the highest final Dice score
- Model B remained competitive but slightly lower

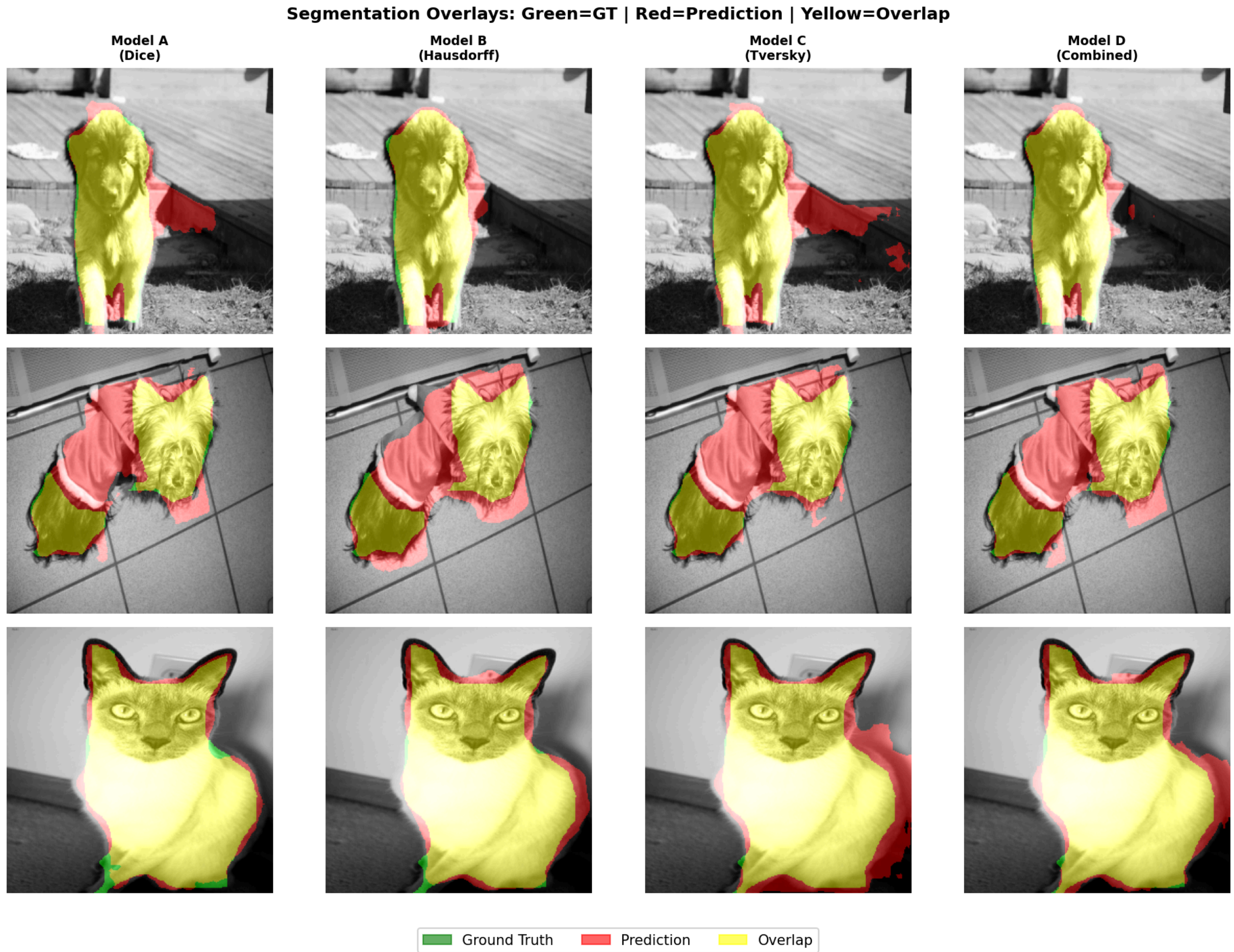
- Model C fluctuated more strongly
- Model D converged smoothly near Model A

The fluctuations in early epochs were caused by limited dataset size and augmentation variability.



7.3 Visual Comparison





The visual comparisons showed that:

- Model A produced strong overall segmentation masks
- Model B produced cleaner object boundaries
- Model C occasionally over-segmented or missed boundaries
- Model D produced balanced masks with stable overlap quality

Overlay visualizations confirmed that Model B aligned boundaries more accurately with ground truth masks.

---

8. Comparative Discussion

Best Overall Performance

Model A (Dice Loss) achieved the best overall segmentation performance.

Reasons:

- Highest Dice Score
- Highest IoU
- Strong convergence stability
- Direct optimization of overlap quality

Best Boundary Accuracy

Model B (Hausdorff-inspired Loss) achieved the best boundary accuracy.

Reason:

- Lowest HD95 value
- Distance-transform weighting penalized boundary errors more effectively

---

Best Region Overlap

Model A produced the best region overlap.

Reason:

- Dice Loss directly optimizes overlap similarity between prediction and target masks

---

Training Stability

- Model A: most stable overall
- Model B: stable with small oscillations
- Model C: most fluctuating
- Model D: stable but slower convergence

Trade-offs Between Overlap-Based and Boundary-Aware Losses

Overlap-Based Losses

(Dice and Tversky)

Advantages:

- Strong region segmentation
- Better Dice and IoU

Disadvantages:

- Less sensitive to contour precision

Boundary-Aware Losses

(Hausdorff-inspired)

Advantages:

- Better boundary localization
- Lower HD95

Disadvantages:

- Slightly weaker overlap metrics

Effectiveness of the Combined Loss

<u>Selected losses:</u>

- A_Dice (Dice Score: `0.8466`)
- C_Tversky (Dice Score: `0.8397`)

These were selected because they achieved the highest validation Dice scores.

The combined loss produced balanced results but did not outperform the best individual models.

Reason:

- Combining Dice and Tversky improved general overlap consistency
- However, both losses optimize similar objectives
- Boundary-specific optimization from Hausdorff loss was not included

Therefore, the combined model improved balance but did not produce the best metric in any category.

---

9. Conclusion

Four semantic segmentation models were trained using different loss functions on the Oxford-IIIT Pet Dataset.

Main findings:

- Dice Loss achieved the best overall segmentation accuracy
- Hausdorff-inspired Loss achieved the best boundary accuracy
- Tversky Loss provided flexible overlap optimization but weaker boundary performance

- The combined loss produced balanced performance but did not exceed the strongest individual models

The experiments demonstrated the trade-off between overlap optimization and boundary-aware optimization in semantic segmentation tasks.